INCH-POUND

MS27399A
w/Amendment 1
21 April 2023
SUPERSEDING
MS27399A
22 March 2005

DETAIL SPECIFICATION SHEET

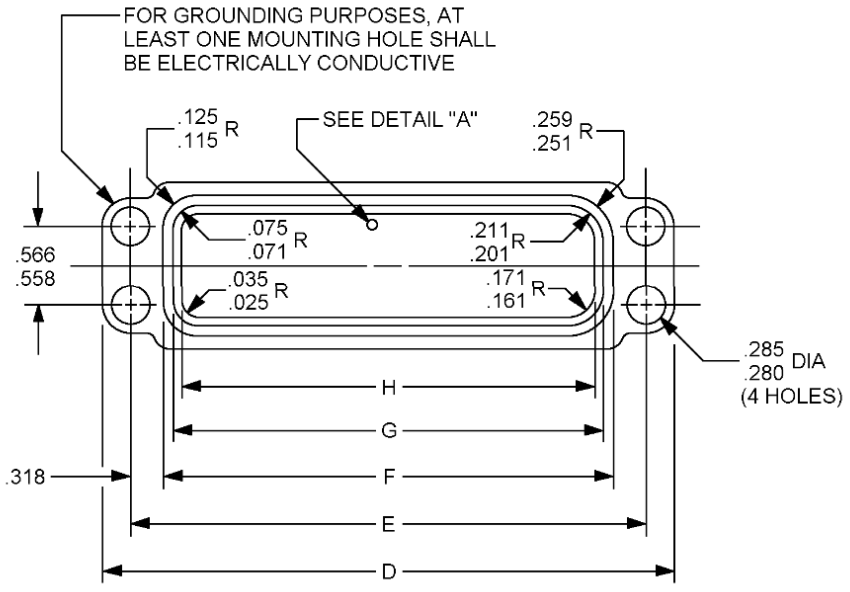
# CONNECTOR, ELECTRICAL, STYLE S, RACK AND PANEL, MINIATURE, CLOSED ENTRY

Inactive for new design after 8 June 1999. For new design use MIL-DTL-83733.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-83733.

The Qualified Products List (QPL) associated with this inactive for new design specification will be maintained until acquisition of the product is no longer required, where upon the specification and QPL will be canceled.




Style S (socket insert)

Environment Resistant (R) Plugs

FIGURE 1. Connector dimensions and configurations.

AMSC N/A FSC 5935

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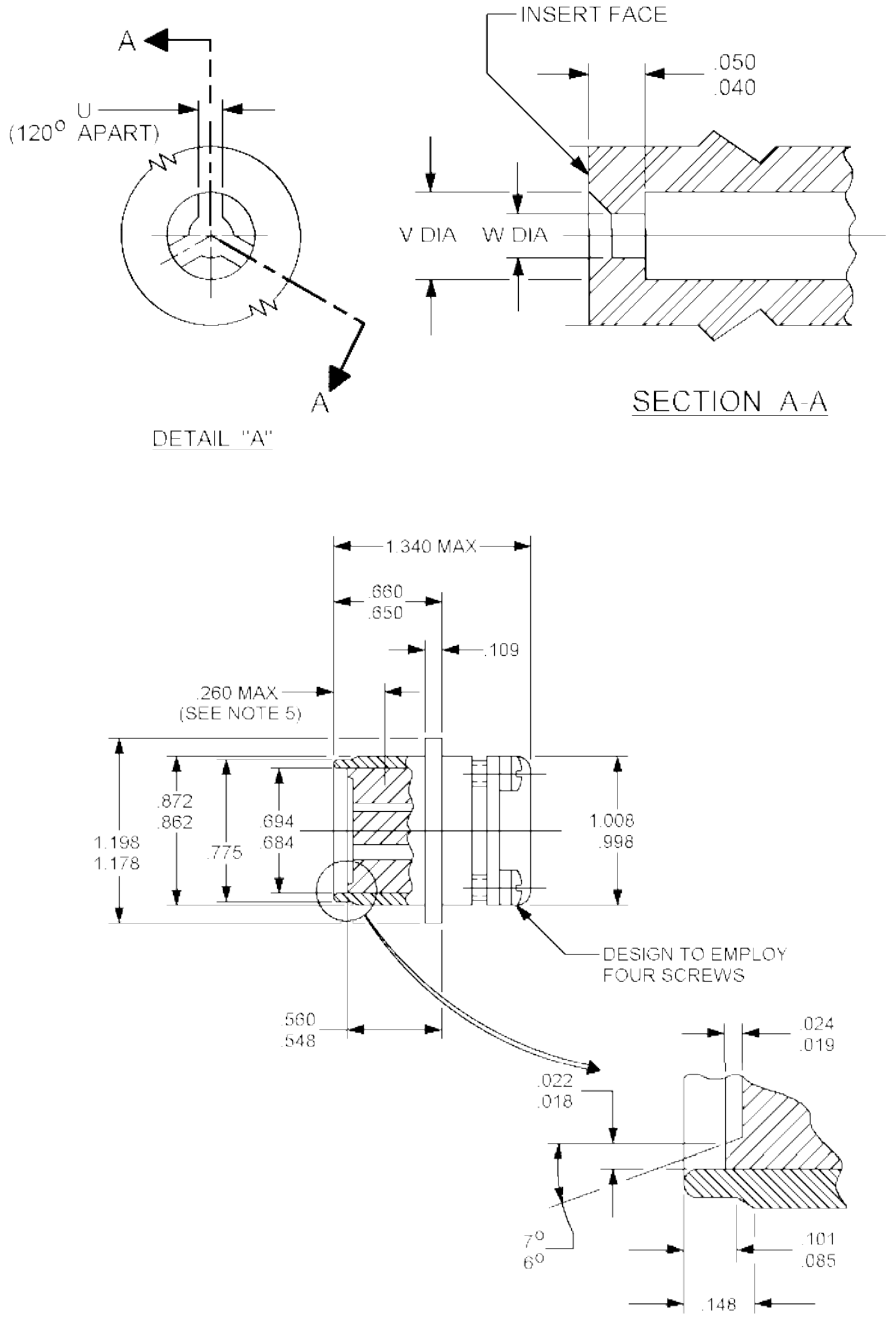



FIGURE 1. Connector dimensions and configurations - Continued.



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| Shell size | D ±.005 | E ±.004 | F ±.005 | G | H ±.005 |
|---|---|---|---|---|---|
| A | 4.296 | 3.875 | 3.239 | 3.147 | 3.061 |
| B | 2.866 | 2.445 | 1.809 | 1.717 | 1.631 |
| C | 2.421 | 2.000 | 1.380 | 1.288 | 1.202 |

Contact pocket dimensions.

| | Contact size | | | | | |
|---|---|---|---|---|---|---|
| | 20 | 16 | 12 | 8 | 4 | Shielded |

| Dimension | | | | | | | | | | | No. 1 | | No. 2 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | min | max | min | max | min | max | min | max | min | max | min | max | min | max |
| U | .017 | .020 | .028 | .031 | .034 | .038 | .040 | .044 | .040 | .044 | .034 | .038 | .040 | .044 |
| V | .110 | .113 | .145 | .148 | .192 | .196 | .283 | .287 | .368 | .372 | .192 | .196 | .283 | .287 |
| W | .048 | .062 | .082 | .086 | .129 | .133 | .224 | .228 | .308 | .312 | .129 | .133 | .224 | .228 |

| Inches | mm | Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|---|---|
| .017 | 0.43 | .085 | 2.16 | .259 | 6.58 | .872 | 22.15 |
| .018 | 0.46 | .086 | 2.18 | .260 | 6.60 | .998 | 25.35 |
| .019 | 0.48 | .101 | 2.57 | .280 | 7.11 | 1.008 | 25.60 |
| .020 | 0.51 | .109 | 2.77 | .283 | 7.19 | 1.178 | 29.92 |
| .022 | 0.56 | .110 | 2.79 | .285 | 7.24 | 1.198 | 30.43 |
| .024 | 0.61 | .113 | 2.87 | .287 | 7.29 | 1.202 | 30.53 |
| .025 | 0.64 | .125 | 3.18 | .308 | 7.82 | 1.288 | 32.72 |
| .028 | 0.71 | .129 | 3.28 | .312 | 7.92 | 1.340 | 34.04 |
| .031 | 0.79 | .133 | 3.38 | .318 | 8.08 | 1.380 | 35.05 |
| .034 | 0.86 | .145 | 3.68 | .368 | 9.35 | 1.631 | 41.43 |
| .035 | 0.89 | .148 | 3.76 | .372 | 9.45 | 1.717 | 43.61 |
| .038 | 0.97 | .161 | 4.09 | .548 | 13.92 | 1.809 | 45.95 |
| .040 | 1.02 | .171 | 4.34 | .558 | 14.17 | 2.000 | 50.80 |
| .044 | 1.12 | .192 | 4.88 | .566 | 14.38 | 2.421 | 61.49 |
| .048 | 1.22 | .196 | 4.99 | .650 | 16.51 | 2.445 | 62.10 |
| .050 | 1.27 | .201 | 5.11 | .660 | 16.76 | 2.866 | 72.80 |
| .062 | 1.57 | .211 | 5.36 | .684 | 17.37 | 3.061 | 77.75 |
| .071 | 1.80 | .224 | 5.69 | .694 | 17.63 | 3.147 | 79.93 |
| .075 | 1.91 | .228 | 5.79 | .775 | 19.69 | 3.239 | 82.27 |
| .082 | 2.08 | .251 | 6.38 | .862 | 21.89 | 3.875 | 98.43 |
| | | | | | | 4.296 | 109.12 |

1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified tolerances are ±.003 inch (0.08 mm).
4. For insert arrangements see MS24288.
5. Distance between end of shell and point at which a gage pin, having the same basic diameter as the mating contact and a square face, engages the socket contact spring.

FIGURE 1. Connector dimensions and configurations - Continued.



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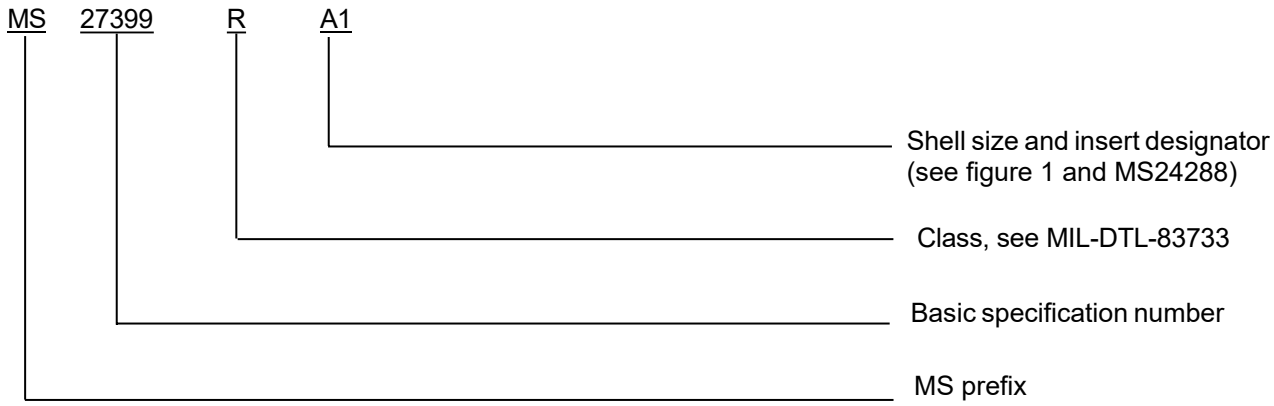
REQUIREMENTS:

Dimensions and configuration: See figure 1.

Materials shall be in accordance with MIL-DTL-26518.

For mating and mounting details see MS24287 and MS24288.

Part or Identifying Number (PIN): See the following example:




Referenced documents. In addition to MIL-DTL-83733, this document references the following:

MS24287
MS24288

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

CONCLUDING MATERIAL

Custodians:
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2023-040)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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